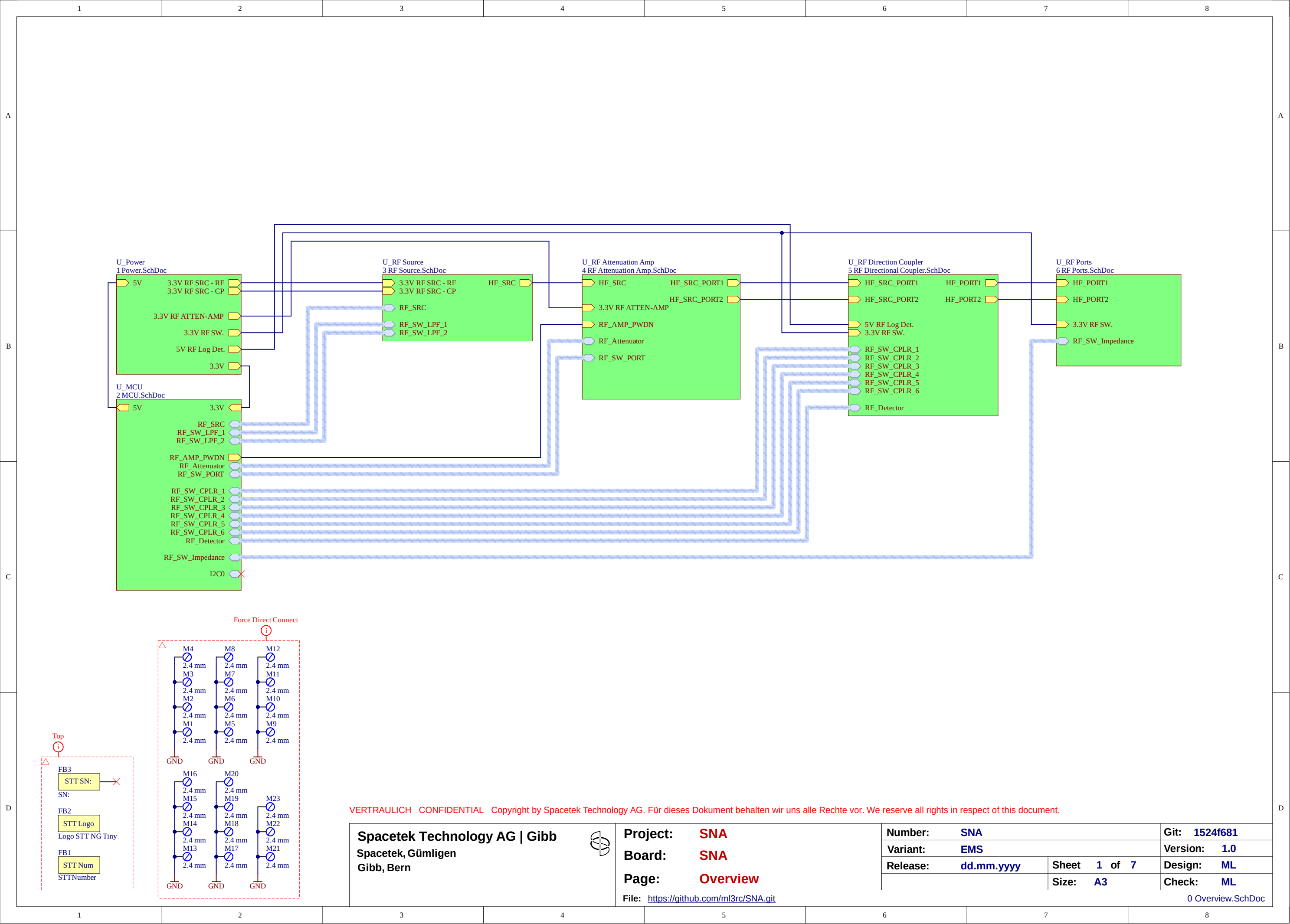
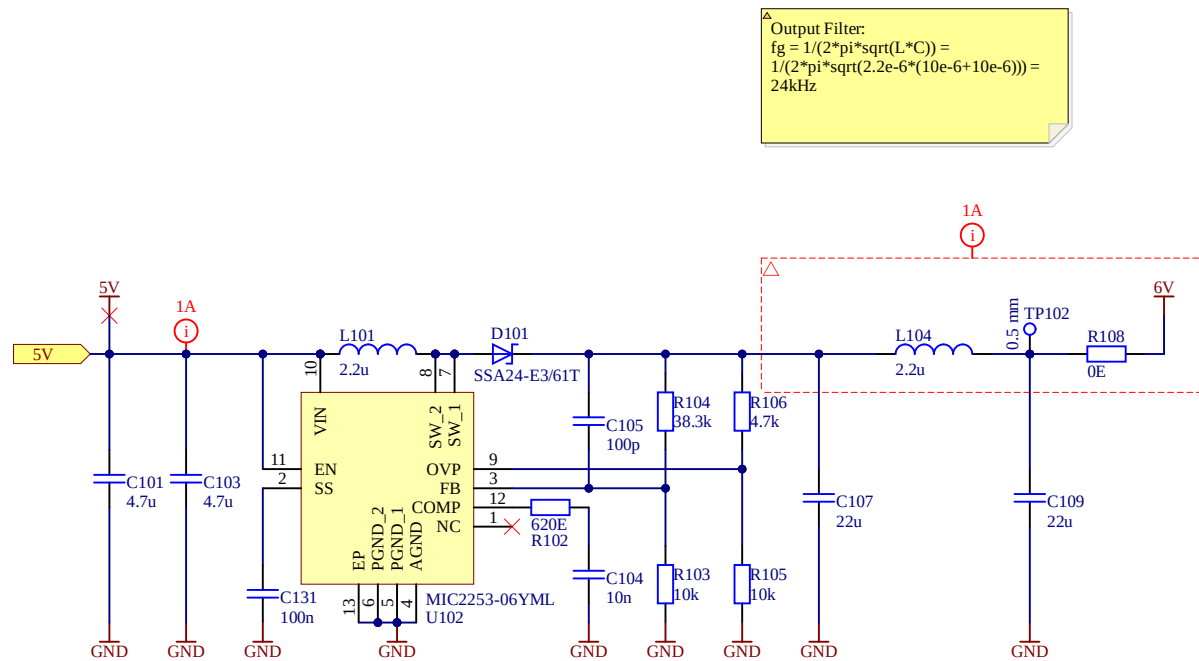
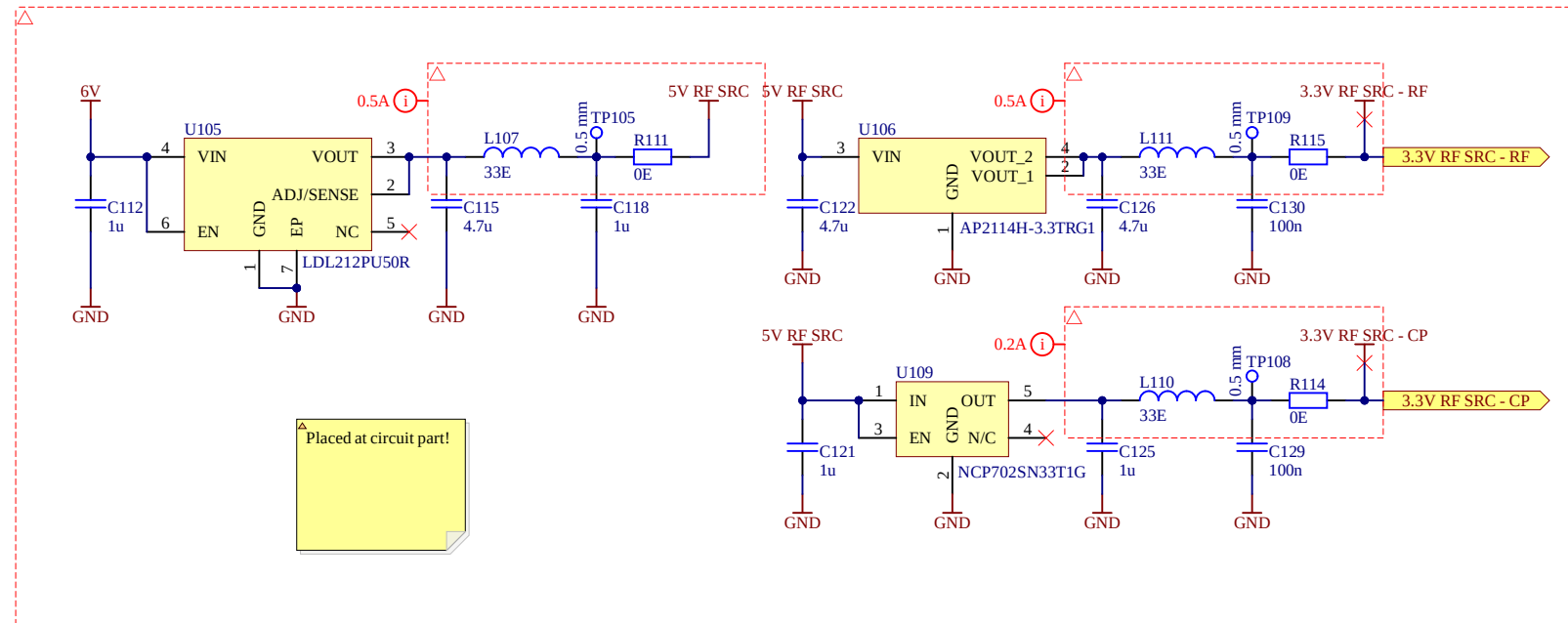






Output Filter:

$$f_g = 1 / (2 * \pi * \sqrt{L * C}) = 1 / (2 * \pi * \sqrt{2.2e-6 * (10e-6 + 10e-6)}) = 24 \text{ kHz}$$



Placed at circuit part!

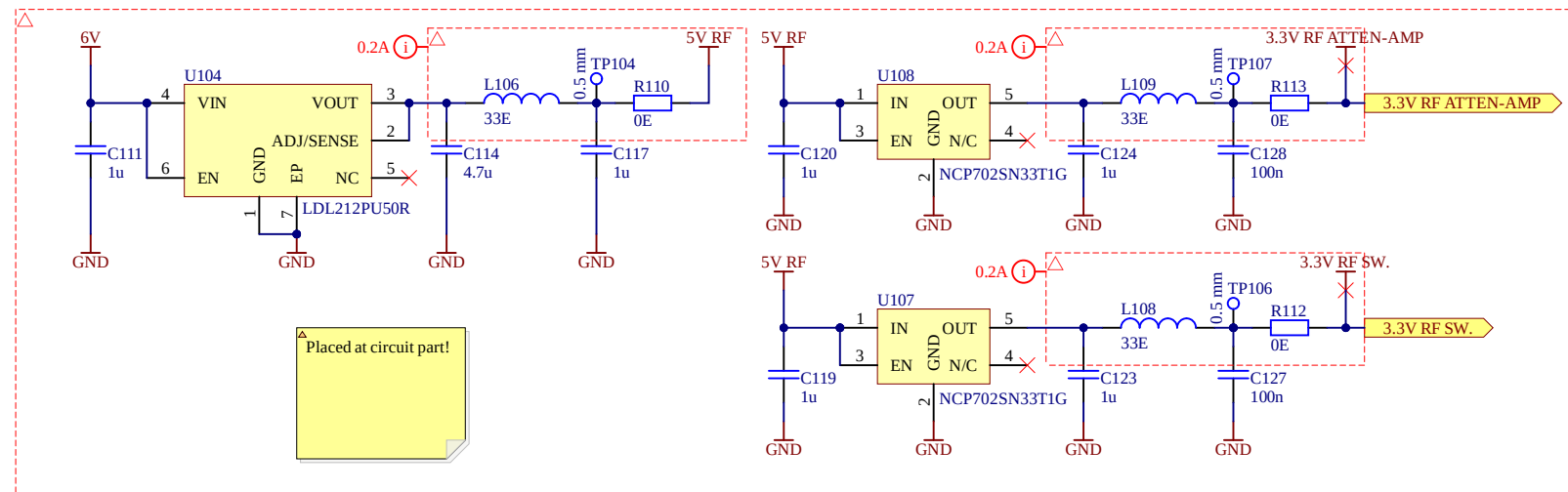
$L = 2.2\mu\text{H}$ as per reference design
 $C_{out} = 22\mu$ as per reference design

 Set ss to 10ms
 $C_{ss} = (T_{ss} - 1\text{ms}) / 85\text{k}\Omega = (10\text{ms} - 1\text{ms}) / 85\text{k}\Omega = 105\text{nF} \rightarrow 100\text{nF}$ (10nF internal cap!)

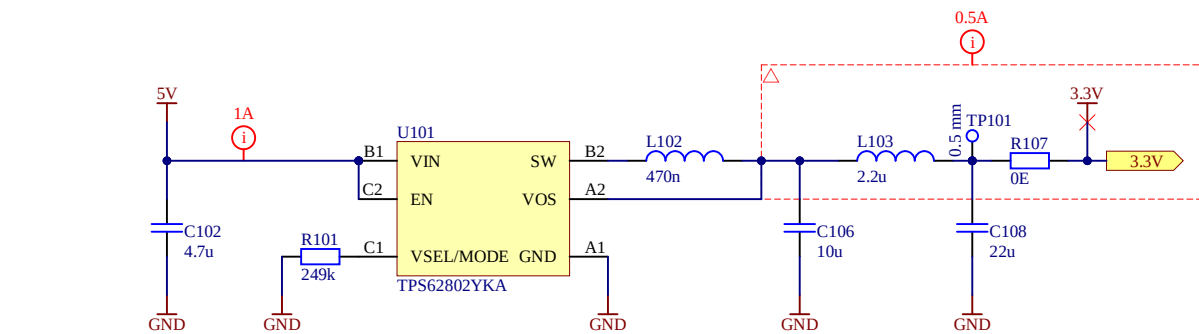
 $U_{ovp} = 8\text{V}$
 $R_{bottom} = 10\text{k}$
 $R_{top} = (U_{ovp} * 15\text{k}\Omega * R_{bottom}) / (1.245 * 67\text{k}\Omega) - R_{bottom} = (8\text{V} * 15\text{k}\Omega * 10\text{k}\Omega) / (1.245 * 67\text{k}\Omega) - 10\text{k}\Omega = 4.39\text{k}\Omega \rightarrow 4.7\text{k}\Omega$

 $U_{out} = 6\text{V}$
 $U_{fb} = 1.245\text{V}$
 $I_{fb} = -450\text{nA}$
 $R_{bottom} = 10\text{k}\Omega$
 $R_{top} = (U_{out} - U_{fb}) / (U_{fb} / R_{bottom} + I_{fb}) = (6\text{V} - 1.245\text{V}) / (1.245\text{V} / 10\text{k}\Omega - 450\text{nA}) = 38.3\text{k}\Omega$

 $C_{comp} = 10\text{nF}$ as per reference design
 $R_{comp} = 620\Omega$ as per reference design



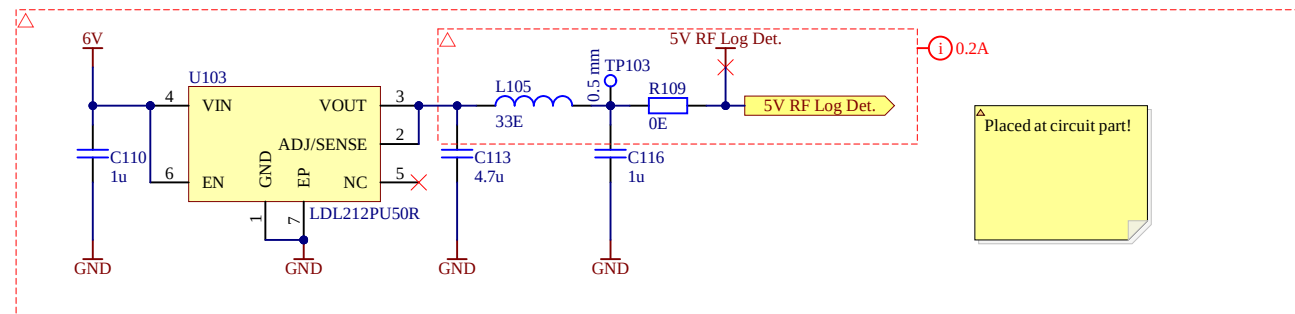
Placed at circuit part!



Design from WEBENCH

Output Filter:

$$f_g = \frac{1}{2\pi \sqrt{L \cdot C}} = \frac{1}{2\pi \sqrt{2.2 \times 10^{-6} \cdot (10 \times 10^{-6} + 10 \times 10^{-6})}} = 24 \text{ kHz}$$



Placed at circuit part!

For current specs see power blockdiagramm
20mil -> 1A (2.14A)
15mil -> 0.5A (1.79A)
12mil -> 0.2A (1.56A)

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Project: SNA
Board: SNA
Page: Power

File: <https://github.com/ml3rc/SNA.git>

Number: SNA

Variant:	EMS
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Release: dd.mm.yyyy

Sheet 2 of 7

Size:	A3
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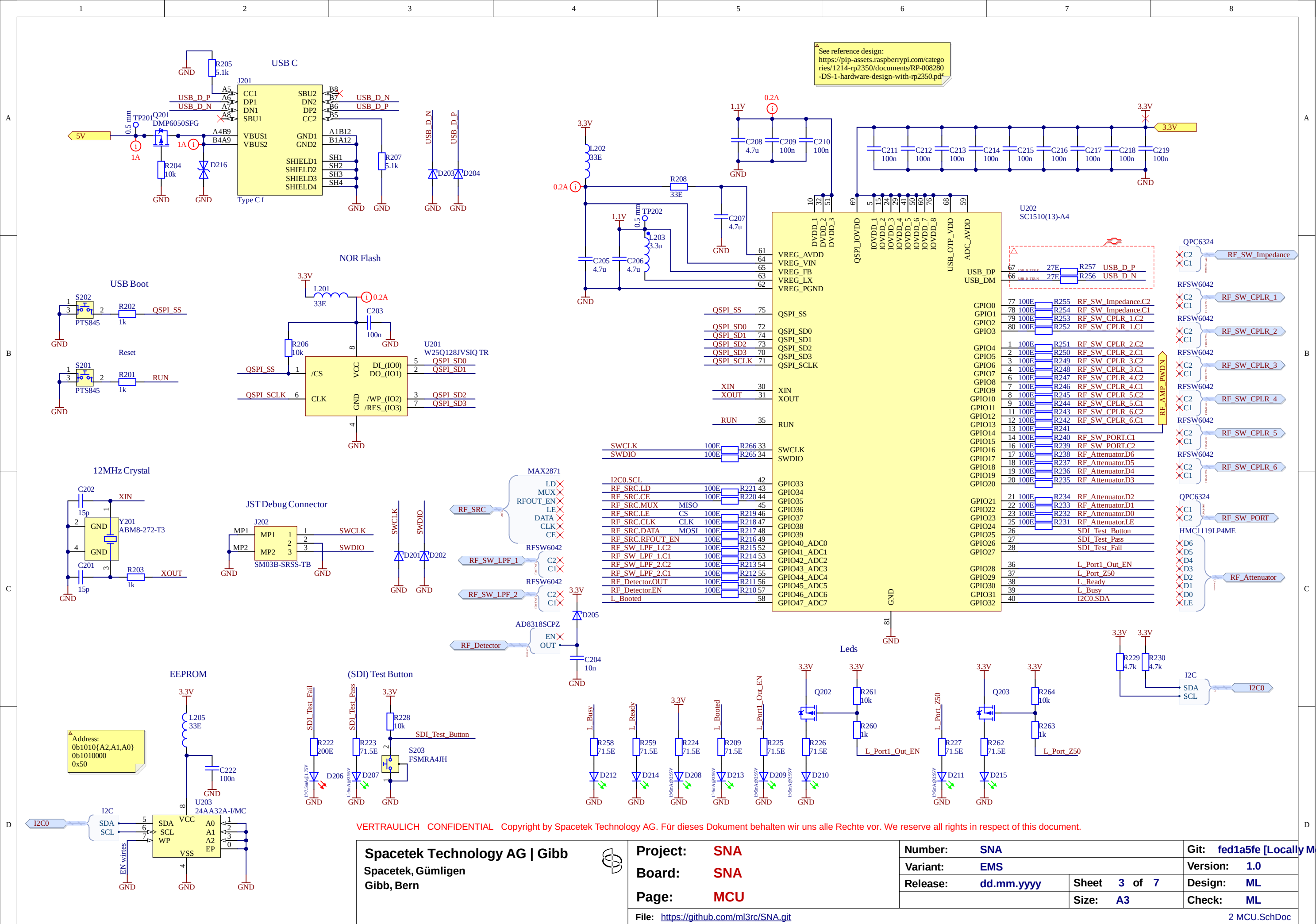
Git: 0553af0a [Locally Modified]

Version:	1.0
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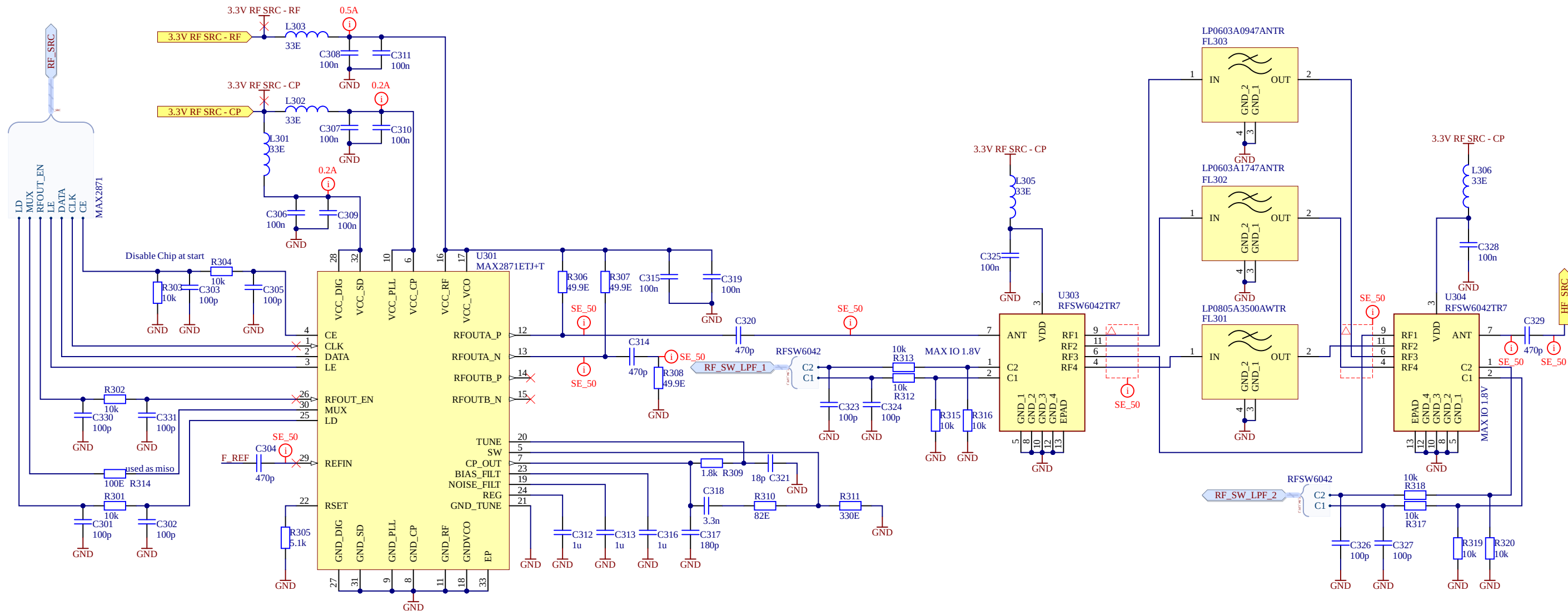
Design:	ML
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Check:	ML
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1 Power.SchDoc



See LibreVNA Schematics for reference



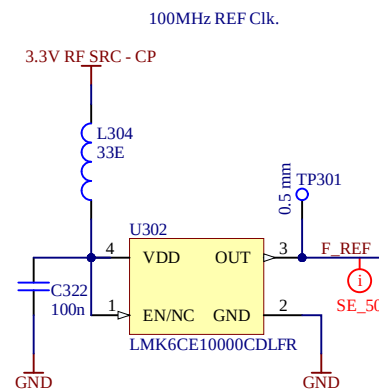
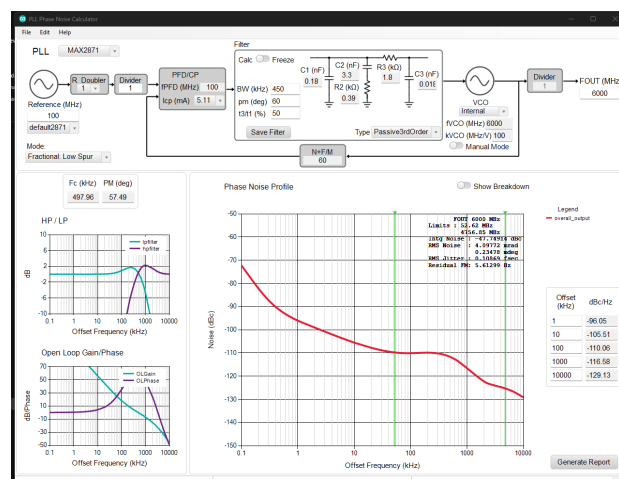
Rset = 5.1kOhm as per datasheet

C_BIAS_FILTER, C_NOISE_FILTER and C_REG = 1uF as per datasheet

The loop filter can be calculated with:
<https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SF0007420A>

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

$R_{crf_coupling} = ((1)/(2 * \pi * 50MHz * 470pF)) = 6.80\Omega$



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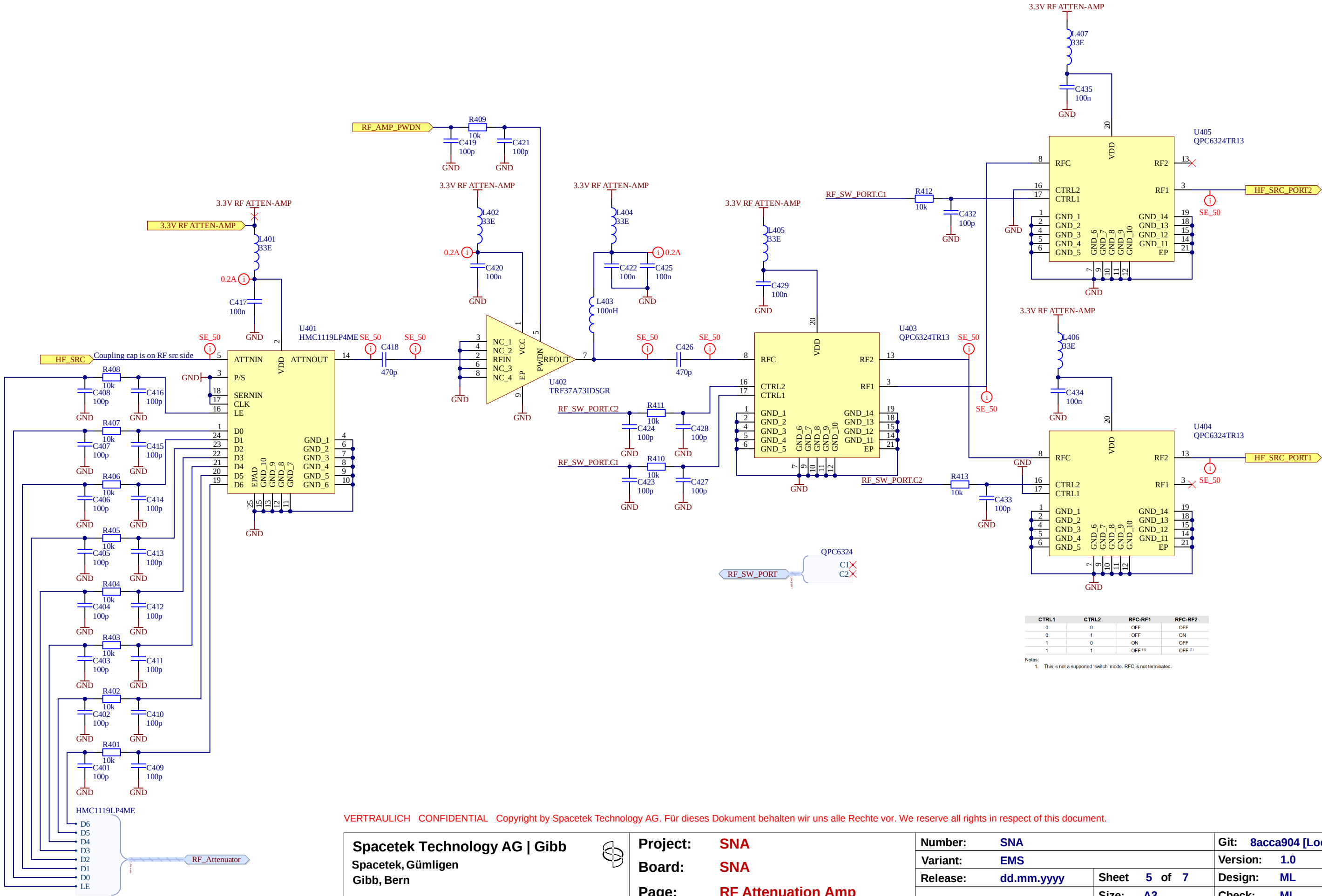


Project: SNA
Board: SNA
Page: RF Source

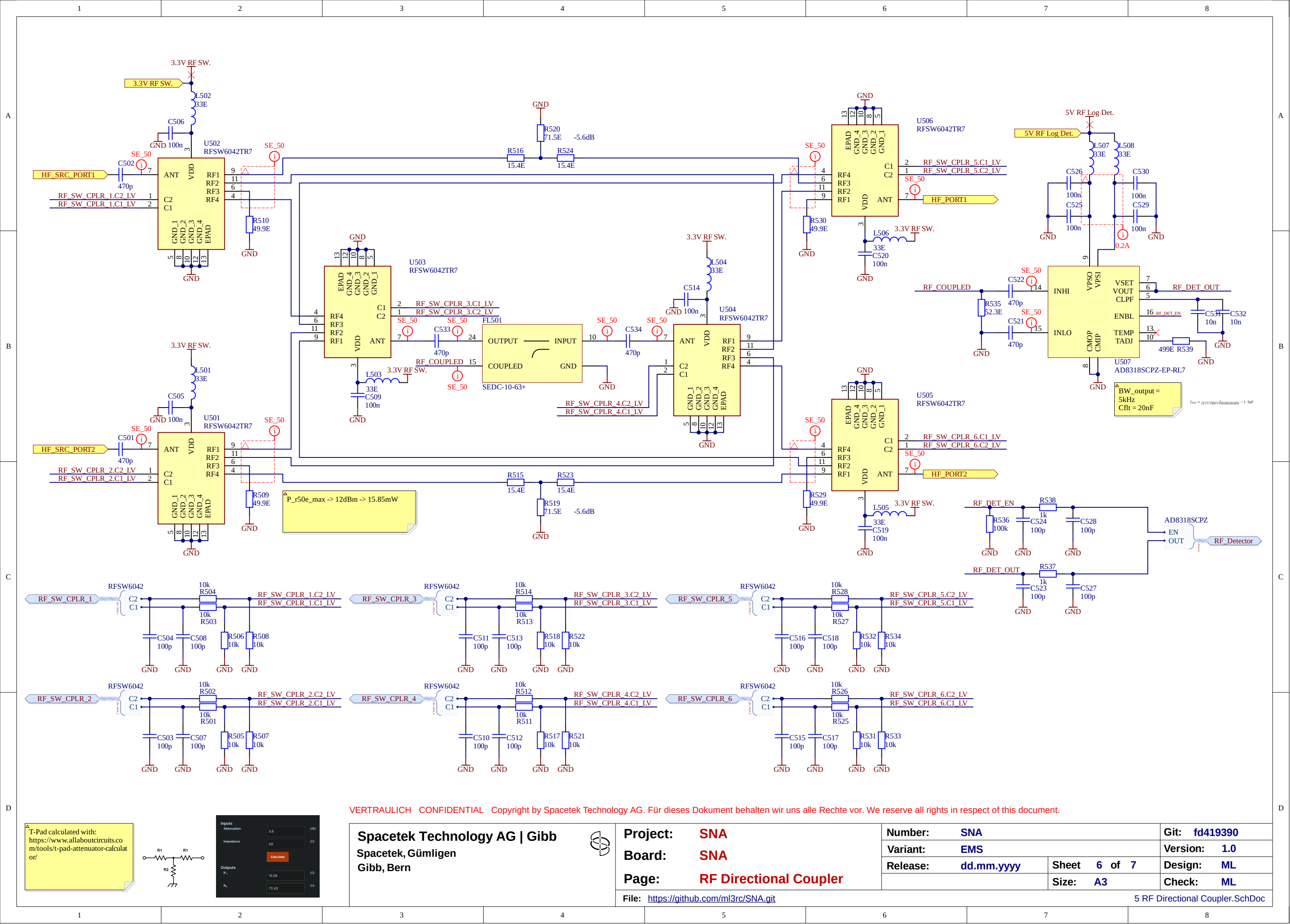
File: <https://github.com/ml3rc/SNA.git>

Number:	SNA	Git:	fd419390
Variant:	EMS	Version:	1.0
Release:	dd.mm.yyyy	Sheet	4 of 7
	Size:	A3	Design: ML
		Check:	ML

3 RF Source.SchDoc



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50-75 Matching Pad:
 $\text{solve}(\{75 = ((1)/(((1)/(R_p)) + ((1)/(50)))) + R_s, 50 = ((1)/(((1)/(75 + R_s)) + ((1)/(R_p))))\})$
 $R_p = 86.6\Omega$
 $R_s = 43.3\Omega$
Insertion Loss = 5.72dB

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Project: **SNA**
Board: **SNA**
Page: **RF Ports**

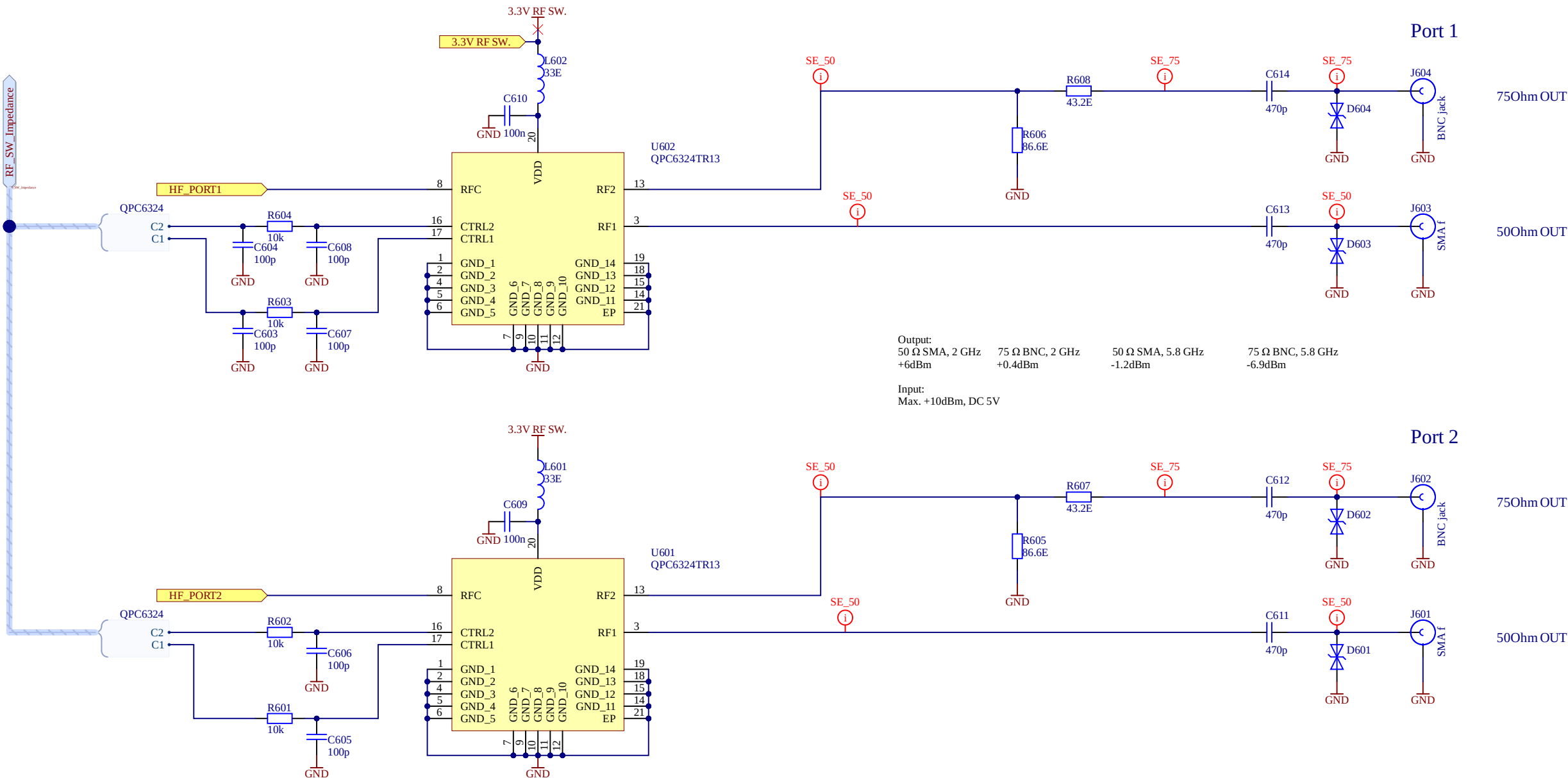
File: <https://github.com/ml3rc/SNA.git>

Number: **SNA**
Variant: **EMS**
Release: **dd.mm.yyyy**

Sheet **7 of 7**
Size: **A3**

Git: **fed1a5fe [Locally M...]**
Version: **1.0**
Design: **ML**
Check: **ML**

6 RF Ports.SchDoc



Output:
50 Ω SMA, 2 GHz
+6dBm

75 Ω BNC, 2 GHz
+0.4dBm

50 Ω SMA, 5.8 GHz
-1.2dBm

75 Ω BNC, 5.8 GHz
-6.9dBm

Input:
Max. +10dBm, DC 5V